

CMPT 354 Mini Project

Step 2: Entity–Relationship Model

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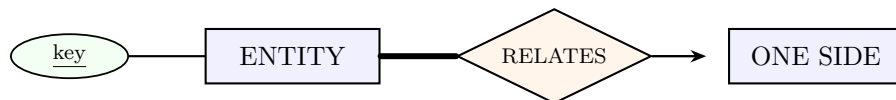
1 Model and Notation

The following E/R model implements the library requirements specified in Step 1. The model separates catalogue descriptions from the individual physical copies and digital licences that can be borrowed. It also represents loans and fines, personnel, events and registrations, future-item proposals, volunteer applications, and help requests.

The diagrams use the notation introduced in class:

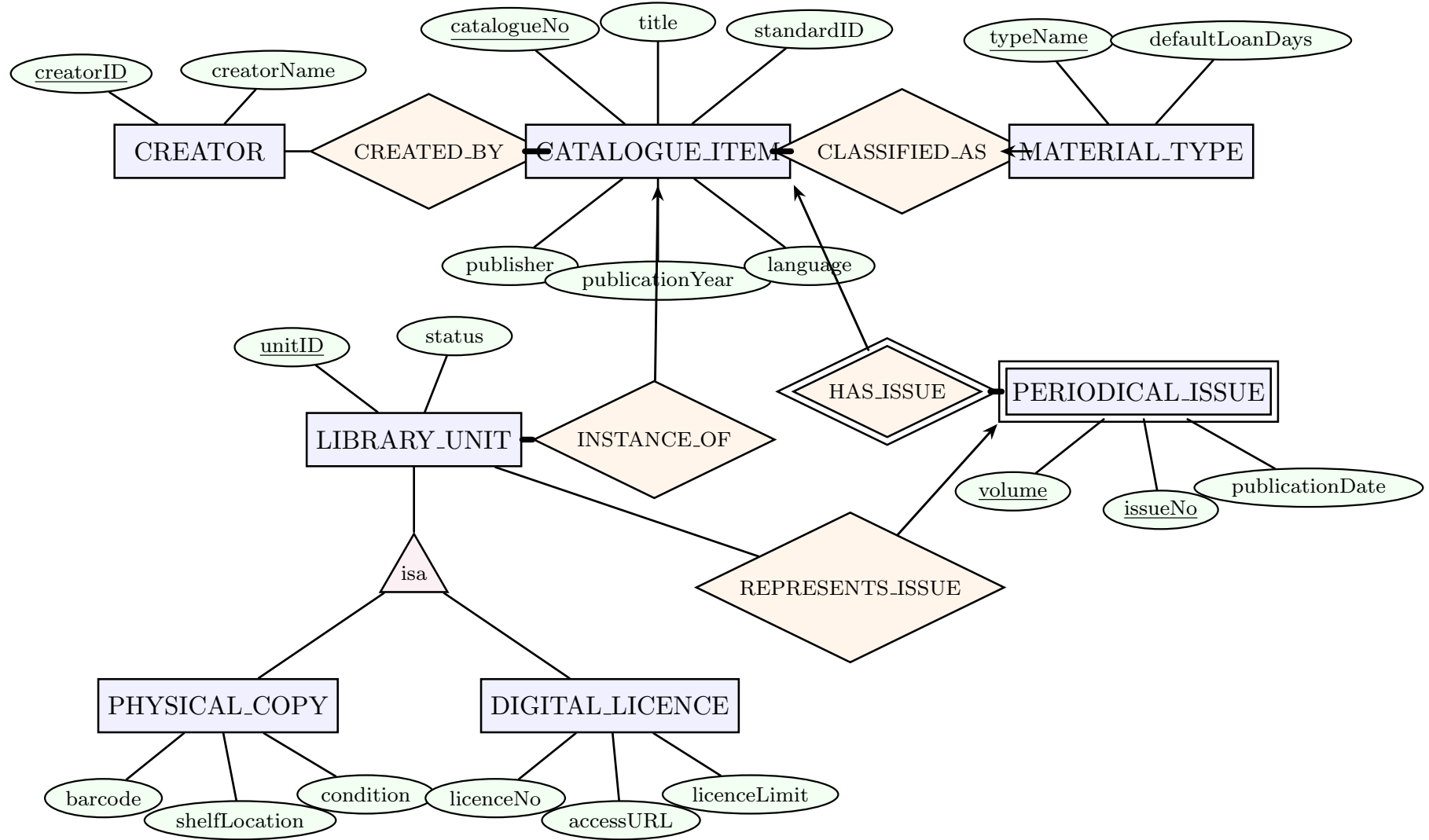
- a rectangle is an entity set, an oval is an attribute, and a diamond is a relationship set;
- an underlined attribute is a primary key;
- an arrow points toward the “one” side of a relationship;
- a thick line represents total participation;
- a triangle represents an *isa* relationship; and
- a double rectangle and double diamond represent a weak entity set and its identifying relationship.

Only the most important attributes are drawn as ovals to keep the diagrams readable. Section 7 gives the complete attribute set and is part of the E/R definition. Attributes marked **AK** there are additional candidate keys, but only the selected primary key is underlined in the diagrams, as required by the class notation.



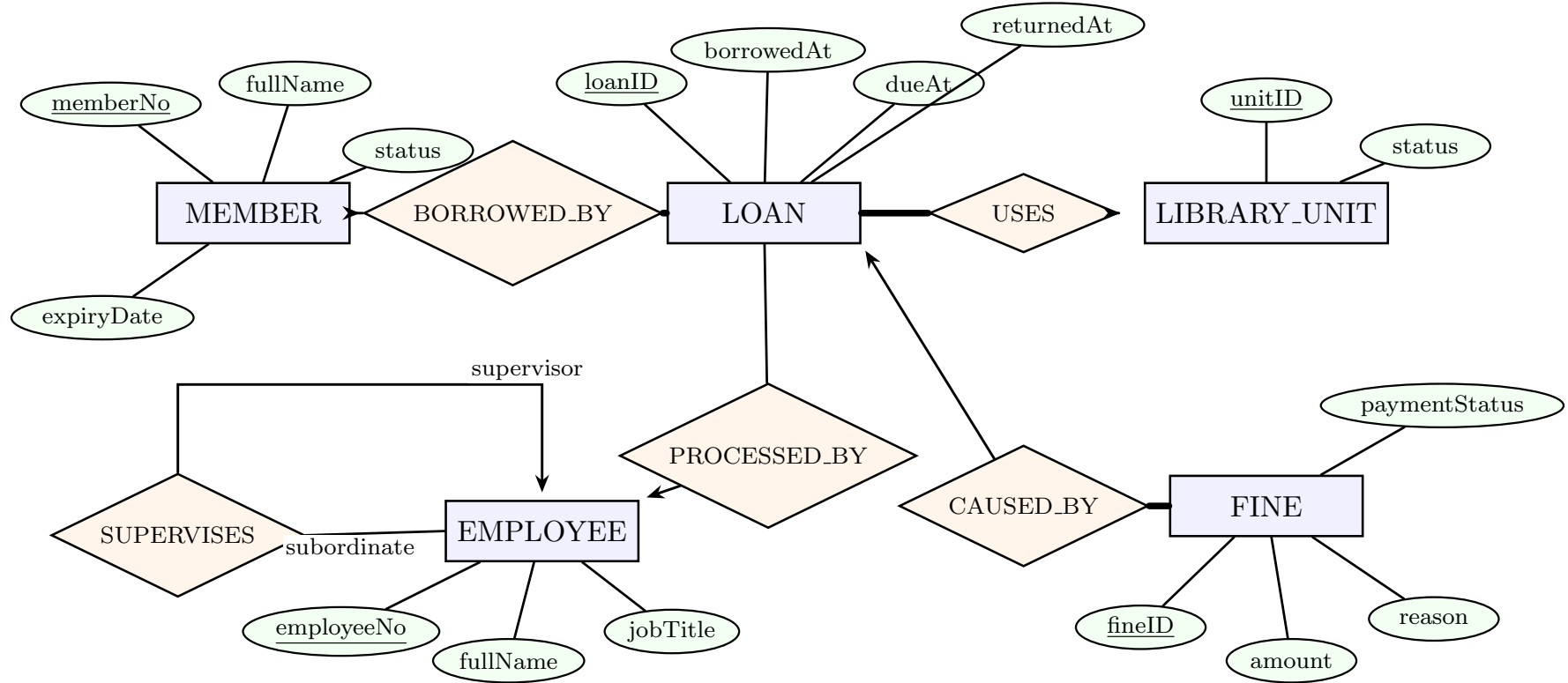
In the example above, every ENTITY participates in exactly one RELATES relationship with ONE SIDE. The thick line gives the minimum of one, and the arrow gives the maximum of one.

2 Catalogue and Lendable Units



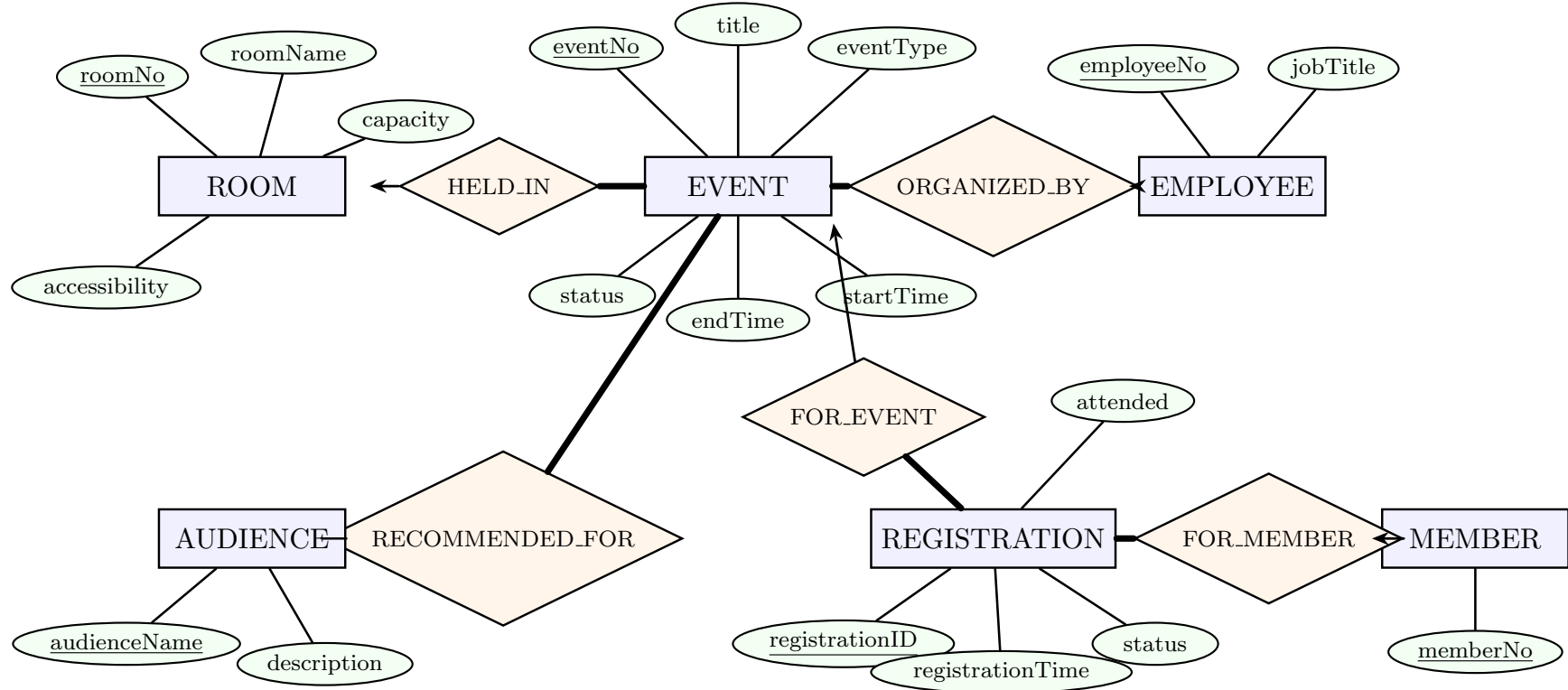
Cardinality and participation. CREATED_BY is many-many, and every catalogue item has at least one creator. Every catalogue item has exactly one material type. Every library unit is an instance of exactly one catalogue item. LIBRARY_UNIT is the superclass of PHYSICAL_COPY and DIGITAL_LICENCE; the subclasses are disjoint and total. PERIODICAL_ISSUE is weak: its key is (*catalogueNo*, *volume*, *issueNo*). REPRESENTS_ISSUE is optional for a unit and is used only for magazine and scientific-journal issues.

3 Members, Personnel, Loans, and Fines



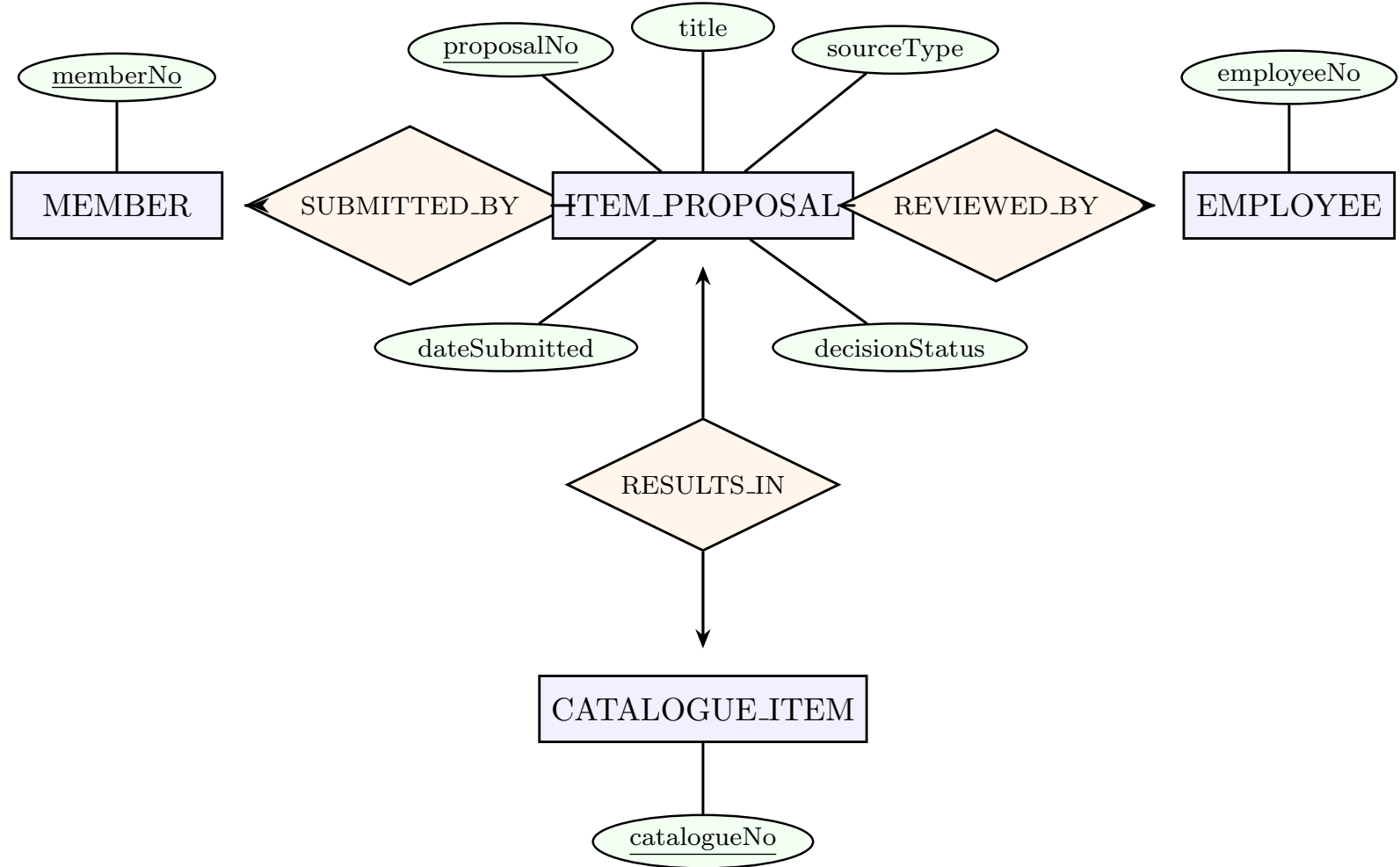
Cardinality and participation. Every loan is for exactly one member and exactly one library unit. A member and a unit may each occur in many loans over time. **PROCESSED_BY** is optional because a digital loan may be completed without an employee. Every fine is caused by exactly one loan, while one loan may cause zero or more fines. In the self-relationship **SUPERVISES**, an employee has at most one supervisor but may supervise many employees.

4 Rooms, Events, Audiences, and Registrations



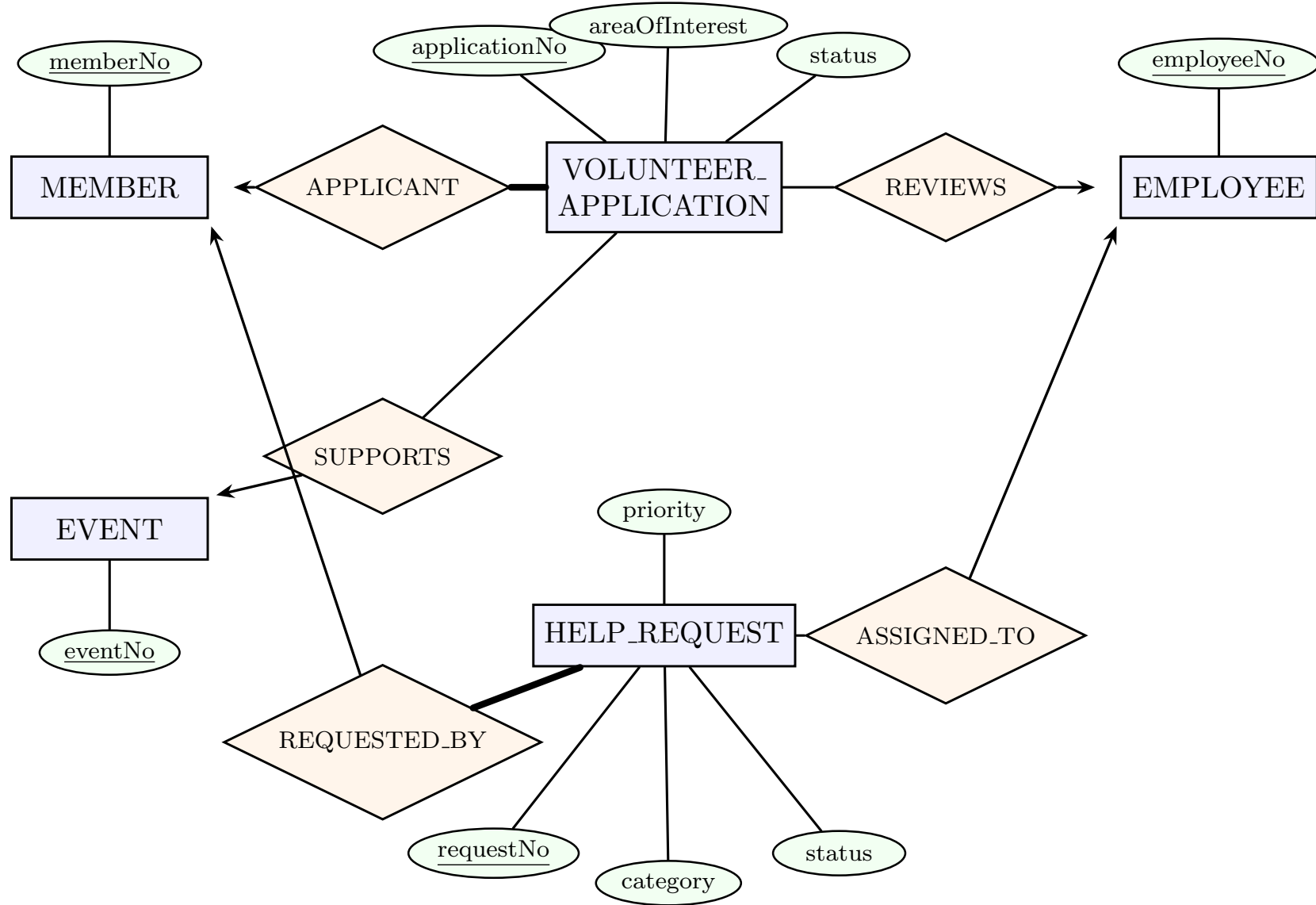
Cardinality and participation. Every event is held in exactly one room and organized by exactly one employee. **RECOMMENDED_FOR** is many-many, and every event has at least one recommended audience. **REGISTRATION** is a connecting entity: every registration belongs to exactly one event and exactly one member. The pair (*eventNo*, *memberNo*) is additionally unique, preventing a member from registering for the same event twice.

5 Future-Item and Donation Proposals



Cardinality and participation. SUBMITTED_BY is optional because personnel may enter a purchase suggestion; it is required when *sourceType* is “member donation” or “member request.” A proposal is reviewed by at most one employee and may result in at most one catalogue item. RESULTS_IN is one-one and optional on both sides.

6 Volunteering and Help Requests



Cardinality and participation. Every volunteer application has exactly one member applicant. Its reviewing employee and supported event are optional, and each is limited to at most one. Every help request is submitted by exactly one member and is assigned to at most one employee.

7 Complete Entity and Attribute Dictionary

Entity set	Attributes
MEMBER	<u>memberNo</u> , fullName, email (AK), phone, mailingAddress, dateJoined, status, expiryDate.
EMPLOYEE	<u>employeeNo</u> , fullName, email (AK), phone, jobTitle, hireDate, employmentStatus.
CREATOR	<u>creatorID</u> , creatorName.
MATERIAL_TYPE	<u>typeName</u> , defaultLoanDays. Examples include print book, online book, magazine, scientific journal, record, and film.
CATALOGUE_ITEM	<u>catalogueNo</u> , title, publisher, publicationYear, language, subject, description, standardID (AK when present).
PERIODICAL_ISSUE	Weak entity with partial-key attributes <u>volume</u> and <u>issueNo</u> , plus publicationDate. Its full key also includes the identifying CATALOGUE_ITEM's catalogueNo.
LIBRARY_UNIT	<u>unitID</u> , status. Status is available, on loan, reserved for processing, damaged, lost, or withdrawn.
PHYSICAL_COPY	Subclass of LIBRARY_UNIT; barcode (AK), shelfLocation, acquisitionDate, condition.
DIGITAL_LICENCE	Subclass of LIBRARY_UNIT; licenceNo (AK), accessURL, licenceLimit.
LOAN	<u>loanID</u> , borrowedAt, dueAt, returnedAt.
FINE	<u>fineID</u> , amount, assessedDate, reason, paymentStatus, paymentDate.
ROOM	<u>roomNo</u> , roomName, floor, capacity, accessibility, availabilityStatus.
EVENT	<u>eventNo</u> , title, eventType, description, startTime, endTime, maximumAttendance, status. Events are free, so no price attribute is needed.
AUDIENCE	<u>audienceName</u> , description.
REGISTRATION	<u>registrationID</u> , registrationTime, status, attended.
ITEM_PROPOSAL	<u>proposalNo</u> , title, creator, materialType, publicationInformation, sourceType, dateSubmitted, condition, notes, decisionStatus, decisionDate.
VOLUNTEER_APPLICATION	<u>applicationNo</u> , applicationDate, areaOfInterest, availability, skills, status, assignmentNotes.
HELP_REQUEST	<u>requestNo</u> , category, questionDescription, submittedAt, priority, status, response, completedAt.

8 Relationship Constraints

Relationship	Multiplicity	Participation rule
CREATED_BY	CATALOGUE_ITEM $M : N$ CREATOR	Every item has at least one creator; a creator may be associated with many items.
CLASSIFIED_AS	CATALOGUE_ITEM $N : 1$ MATERIAL_TYPE	Every item has exactly one type.
HAS_ISSUE	PERIODICAL_ISSUE $N : 1$ CATALOGUE_ITEM	Identifying relationship; every issue belongs to exactly one periodical catalogue item.
INSTANCE_OF	LIBRARY_UNIT $N : 1$ CATALOGUE_ITEM	Every unit is an instance of exactly one item.
REPRESENTS_ISSUE	LIBRARY_UNIT $N : 1$ PERIODICAL_ISSUE	Optional on the unit side; used only for periodical issues.
BORROWED_BY	LOAN $N : 1$ MEMBER	Every loan has exactly one member.
USES	LOAN $N : 1$ LIBRARY_UNIT	Every loan uses exactly one unit.

Relationship	Multiplicity	Participation rule
PROCESSED_BY	LOAN $N : 1$ EMPLOYEE	At most one employee processes a loan; participation is optional.
CAUSED_BY	FINE $N : 1$ LOAN	Every fine is caused by exactly one loan.
SUPERVISES	EMPLOYEE $N : 1$ EMPLOYEE	An employee has at most one supervisor.
HELD_IN	EVENT $N : 1$ ROOM	Every event is held in exactly one room.
ORGANIZED_BY	EVENT $N : 1$ EMPLOYEE	Every event has exactly one organizer.
RECOMMENDED_FOR	EVENT $M : N$ AUDIENCE	Every event has at least one recommended audience.
FOR_EVENT	REGISTRATION $N : 1$ EVENT	Every registration has exactly one event.
FOR_MEMBER	REGISTRATION $N : 1$ MEMBER	Every registration has exactly one member.
SUBMITTED_BY	ITEM_PROPOSAL $N : 1$ MEMBER	Optional generally; required for a member donation or member request.
REVIEWED_BY	ITEM_PROPOSAL $N : 1$ EMPLOYEE	At most one reviewing employee.
RESULTS_IN	ITEM_PROPOSAL $1 : 1$ CATALOGUE_ITEM	Optional; used only after an accepted proposal has been catalogued.
APPLICANT	VOLUNTEER_APPLICATION $N : 1$ MEMBER	Every application has exactly one applicant.
REVIEWS	VOLUNTEER_APPLICATION $N : 1$ EMPLOYEE	At most one employee reviews an application.
SUPPORTS	VOLUNTEER_APPLICATION $N : 1$ EVENT	Optional; absent for general library work.
REQUESTED_BY	HELP_REQUEST $N : 1$ MEMBER	Every request has exactly one requesting member.
ASSIGNED_TO	HELP_REQUEST $N : 1$ EMPLOYEE	At most one employee is assigned.

9 Additional Semantic Constraints

Some business rules depend on dates, counts, sums, or status values and cannot be fully represented by arrows and participation lines in an E/R diagram. They remain part of the conceptual design and will later be enforced with SQL constraints, triggers, transactions, or application logic:

- Only an active member may borrow. A member may have at most five active loans and may not borrow while an item is overdue or unpaid fines total at least \$20.
- A physical copy may have at most one active loan. Concurrent digital loans for a digital licence may not exceed its licenceLimit.
- dueAt must be later than borrowedAt. A late physical return is charged \$0.25 per day, and all fine amounts are nonnegative.
- An event's endTime must be after its startTime. maximumAttendance may not exceed its room's capacity, and two events may not overlap in the same room.
- The pair (memberNo,eventNo) is unique in REGISTRATION. Registration is allowed only for a scheduled event with available capacity.
- A proposal can participate in RESULTS_IN only after it is accepted. Rejected proposals remain stored as history.

- A volunteer application can receive an assignment only after approval. A help request can become resolved only when response and completedAt are present.

10 Coverage of Required Application Actions

The model supports all eight required actions: catalogue and unit records allow item search; LOAN and FINE support borrowing and returning; ITEM_PROPOSAL supports donations and future acquisitions; EVENT, AUDIENCE, ROOM, and REGISTRATION support event search and registration; VOLUNTEER_APPLICATION supports volunteering; and HELP_REQUEST supports asking a librarian for help.